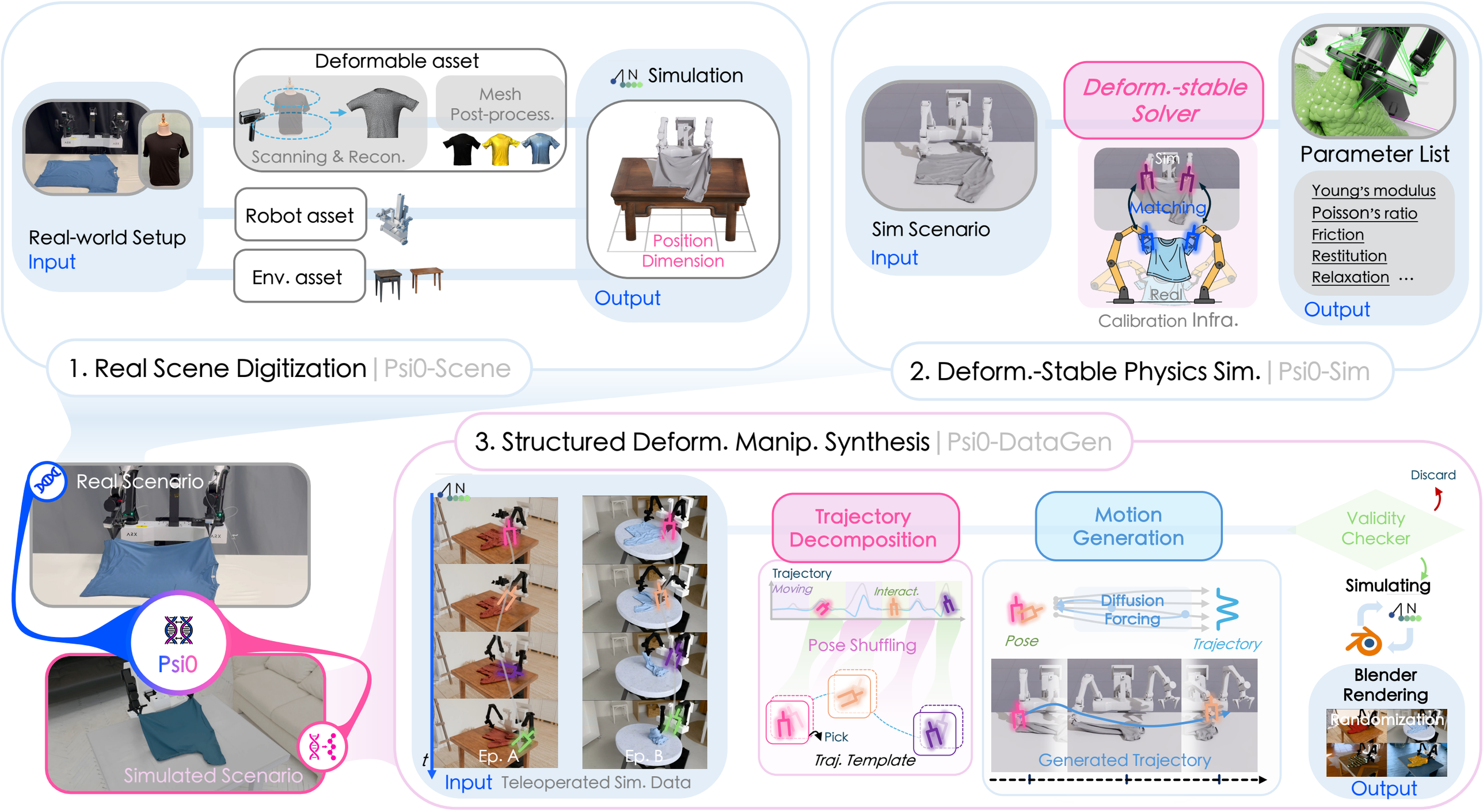
Real-world Setup
Input
Deformable asset
Scanning & Recon.
Mesh Post-process.
Robot asset
Env. asset
Simulation
Position
Dimension
Output
1. Real Scene Digitization | Psi0-Scene
Sim Scenario
Input
Deform.-stable Solver
Sim
Matching
Real
Calibration Infra.
Parameter List
Young's modulus
Poisson's ratio
Friction
Restitution
Relaxation ...
Output
2. Deform.-Stable Physics Sim. | Psi0-Sim
3. Structured Deform. Manip. Synthesis | Psi0-DataGen
Real Scenario
Psi0
Simulated Scenario
Ep. A
Ep. B
t
Input Teleoperated Sim. Data
Trajectory Decomposition
Trajectory
Moving
Interact.
Pose Shuffling
Pick
Traj. Template
Motion Generation
Diffusion Forcing
Pose
Trajectory
Generated Trajectory
Discard
Validity Checker
Simulating
Blender Rendering
Randomization
Output